

## SOFTWARE ENGINEERING

# PROJECT PROPOSAL

## Smart Complaint Management System

|              |                                             |
|--------------|---------------------------------------------|
| Group Number | 8                                           |
| Project Type | Self-contained academic web-based prototype |
| Technology   | Python • Flask • MySQL                      |
| Version      | 1.0                                         |

### Team Members

| Name          | SRN           |
|---------------|---------------|
| Shashank SK   | PES2UG24CS460 |
| Shreya Shenoy | PES2UG24CS487 |
| Sanjan M      | PES2UG24CS442 |
| Samartha MS   | PES2UG24CS435 |

Prepared in accordance with the Software Engineering Synopsis/Project Proposal requirements and the Group 8 SRS and requirements list.

# 1. Project Title and Team Profile

Project Title: Smart Complaint Management System

Team Number: Group 8

The team consists of four members. In accordance with the proposal requirements, every member will participate in all project activities, while individual feature ownership is defined later in this document.

## 2. Proposed Project Description

The Smart Complaint Management System is a new, self-contained, web-based academic prototype for lodging, managing, tracking, escalating, and resolving complaints through a centralized platform. The system is intended to replace fragmented or manual complaint-handling processes within the scope of the prototype by maintaining complaint information and its progress in one place.

A registered user can create an account, lodge a complaint, choose a complaint category, track the complaint, receive relevant notifications, search and filter accessible complaints, and escalate a submitted complaint when it has not received sufficient attention or appropriate action. Authorized administrators can manage complaints, assign them to appropriate personnel or departments, update status, resolve complaints, manage escalations, monitor activities through an administrative dashboard, and view reports and analytics.

The system also includes smart complaint classification. Submitted complaints are analyzed to determine a priority level, helping administrators identify issues requiring greater attention. The system additionally compares relevant complaint information with existing complaints to identify possible duplicates. Possible duplicate relationships are presented for administrator review rather than automatically deleting or merging user submissions.

The proposed system follows a client-server architecture. Users and administrators interact through a browser-based interface; the Flask backend handles application logic and communication with the database; and MySQL provides persistent storage for user, complaint, category, assignment, status, escalation, notification, and related data.

### Technology and environment

- Python is the primary programming language.
- Flask is the web application framework for the backend.
- MySQL is the database management system.
- The application is intended for modern desktop/laptop browsers such as Google Chrome, Mozilla Firefox, and Microsoft Edge.
- HTTPS is preferred for deployment to protect transmitted credentials and complaint information.

### Intended users

- Users: registered individuals who lodge, categorize, track, search/filter, receive notifications for, and escalate their own complaints.
- Administrators: authorized personnel who manage complaints, assignments, statuses, escalations, resolutions, dashboards, reports, and analytics.

### Customer / stakeholder position

No external customer or government/municipal integration is identified in the provided project materials. The system is explicitly scoped as a standalone academic prototype and is not an official government or municipal complaint platform. Future integrations and advanced features such as

multilingual support, voice-based complaint filing, chatbot assistance, or external organizational systems are outside the current project scope.

### 3. Functional Features

At the conclusion of the project, the system is planned to provide the following user-visible functional features:

| No. | Functional Feature                   | Planned Capability                                                                         |
|-----|--------------------------------------|--------------------------------------------------------------------------------------------|
| 1   | User registration and login          | Account creation, authentication, protected access, and role-based entry.                  |
| 2   | Complaint registration               | Submission of complaint details with required validation and confirmation.                 |
| 3   | Complaint categorization             | Selection and maintenance of predefined complaint categories.                              |
| 4   | Complaint tracking and notifications | Viewing complaint status/progress and receiving relevant updates.                          |
| 5   | Complaint assignment                 | Administrator assignment or reassignment to appropriate personnel/department.              |
| 6   | Complaint escalation                 | User escalation of a submitted complaint and administrator review/action.                  |
| 7   | Smart complaint classification       | Automatic priority categorization based on available complaint information.                |
| 8   | Duplicate complaint detection        | Identification of possible duplicate complaints for administrator review.                  |
| 9   | Complaint status updates             | Authorized administrator updates to predefined complaint statuses.                         |
| 10  | Admin dashboard                      | Centralized administrative monitoring and complaint-management interface.                  |
| 11  | Reports and analytics                | Complaint counts, status information, category-based views, and reports.                   |
| 12  | Role-based access control            | Restriction of functions and information according to authorized role.                     |
| 13  | Search and filtering                 | Search and filtering of complaints using available attributes such as category and status. |

## 4. Plan of Work - Short-Term Development Plan

The following short-term plan translates the provided requirements into a practical sequence for the next few weeks. The stages are planned so that core complaint handling is established before smart features, administrative functions, integration, and final validation are completed.

| Period | Focus                                      | Planned Work                                                                                                                                                                                                                                                                | Participation                                           |
|--------|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| Week 1 | Foundation and data model                  | Finalize system structure; establish Flask project and MySQL database; define core entities and relationships; implement initial registration/login and role handling; agree on common interfaces and coding conventions.                                                   | All members                                             |
| Week 2 | Core complaint workflow                    | Implement complaint registration, categorization, complaint storage, complaint tracking, predefined statuses, and the basic notification flow. Connect the user workflow end-to-end.                                                                                        | All members; feature owners lead their assigned modules |
| Week 3 | Administrative workflow and smart features | Implement complaint assignment, escalation, status management, admin dashboard, search/filtering, reports/analytics, priority classification, and duplicate complaint detection.                                                                                            | All members; feature owners lead their assigned modules |
| Week 4 | Integration, quality and demonstration     | Integrate all modules; verify role-based restrictions; validate data integrity; test smart-feature accuracy; test usability, reliability, compatibility, accessibility, auditability, backup/recovery, and fault handling; fix defects and prepare the final demonstration. | All members                                             |

## 5. Product Ownership and Team Responsibilities

Product ownership is divided by functional feature so that each member has clear responsibility for delivering a meaningful part of the product. The allocations below are the team's proposed primary ownership plan; all members will still participate in integration, reviews, testing, documentation, and other project activities as required.

| Member        | Primary Functional Ownership                                                                             | Qualitative / Quality Ownership                                                                                                                                                                                             |
|---------------|----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Shashank SK   | User registration and login; Role-based access control; Complaint registration; Complaint categorization | Security and privacy: protect authentication credentials, enforce authorization checks, and limit exposure of personal/complaint information. Maintain clear validation and error handling for account and complaint entry. |
| Shreya Shenoy | Complaint tracking and notifications; Complaint escalation; Complaint status updates                     | Usability and reliability: provide clear complaint progress/status information, understandable notifications, graceful handling of invalid inputs, and reliable reflection of status changes.                               |

| Member      | Primary Functional Ownership                                                                                  | Qualitative / Quality Ownership                                                                                                                                                                        |
|-------------|---------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sanjan M    | Smart complaint classification (priority categorization); Duplicate complaint detection; Search and filtering | Accuracy and performance: evaluate priority/duplicate-detection results during testing and keep search/filter operations responsive for the expected prototype workload.                               |
| Samartha MS | Complaint assignment; Admin dashboard; Reports and analytics                                                  | Data integrity and auditability: preserve consistent assignment/status/reporting data and maintain traceable records of important administrative actions. Support maintainable administrative modules. |

## 6. Shared Team Responsibilities

- Every team member participates in all project activities, including requirement clarification, design discussions, integration, reviews, testing, defect fixing, documentation, and final demonstration.
- Database interaction, user-interface implementation, graphic design, unit testing, and similar engineering activities are treated as supporting responsibilities rather than standalone functional features, consistent with the proposal instructions.
- The team will coordinate module interfaces so that separately owned functionality works as one complaint lifecycle: authentication → registration → categorization/classification → storage → assignment → status updates → escalation/resolution → notification and closure.
- The final system will be evaluated against the functional and non-functional requirements identified in the SRS and requirements list.

## 7. Non-Functional and Quality Goals

The requirements list identifies the following non-functional properties, which will guide implementation and evaluation:

| Quality Property       | Project Goal                                                                                                                           |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Performance</b>     | Responsive complaint registration, tracking, dashboard, search/filtering, and status updates under the expected prototype workload.    |
| <b>Availability</b>    | System should be functional during scheduled testing, demonstration, and evaluation.                                                   |
| <b>Security</b>        | Authentication, protected credentials, HTTPS during deployment, role-based access, and restricted personal/complaint data.             |
| <b>Reliability</b>     | Graceful handling of invalid inputs and unexpected application conditions without loss of valid complaint information.                 |
| <b>Scalability</b>     | Design should allow extension beyond the current prototype scale.                                                                      |
| <b>Usability</b>       | Simple, clear, consistent navigation and understandable complaint/status information.                                                  |
| <b>Maintainability</b> | Structured application, database, and interface components that can be modified without unnecessary impact on unrelated functionality. |
| <b>Data integrity</b>  | Accurate and consistent user, complaint, category, assignment, status, escalation, and notification information.                       |
| <b>Privacy</b>         | Personal and complaint-related information accessible only according to authorization.                                                 |

| Quality Property    | Project Goal                                                                                           |
|---------------------|--------------------------------------------------------------------------------------------------------|
| Backup and recovery | Appropriate procedures should reduce the risk of loss of complaint and user data.                      |
| Compatibility       | Support for commonly used modern browsers, including Chrome, Firefox, and Edge.                        |
| Auditability        | Records of important assignments, status updates, and escalations for traceability and accountability. |
| Accuracy            | Sufficiently accurate smart priority classification and duplicate detection during testing.            |
| Fault tolerance     | Application/input errors handled gracefully with appropriate error messages.                           |
| Accessibility       | Readable text, clear labels, understandable navigation, and appropriately designed controls.           |

## 8. Scope Boundaries

- The system is a standalone academic prototype and does not depend on external government, municipal, or third-party authentication systems.
- No hardware interface is planned for this project.
- Emergency situations and serious crimes requiring immediate intervention are outside the system's scope and should be reported to appropriate emergency or law-enforcement services.
- Large-scale nationwide deployment is outside the current proof-of-concept scope.
- Future enhancements such as external municipal integrations, multilingual support, voice-based filing, chatbot assistance, and advanced communication channels are not part of the current project.

## 9. Reference Documents Used

1. Software Engineering - Synopsis/Project Proposal, provided project proposal instructions. Defines the mandatory proposal sections, team profile, project description, functional feature list, short-term plan of work, functional ownership, and qualitative property ownership.
2. SRS\_g8.pdf - Jackfruit Phase-1 Software Requirements Specification for Smart Complaint Management System, Version 1.0. Provides the project purpose, scope, architecture/technology, user classes, system functions, analysis models, detailed functional requirements, non-functional requirements, business/domain requirements, and scope boundaries.
3. SE\_PROJECT\_G8.pdf - Smart Complaint Management System, Team 8. Provides the confirmed team number, team members/SRNs, and consolidated functional and non-functional requirements list.